

Phase 3 Normalization

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A construction company constructs houses for a number of clients. A client can have a number of houses/offices built by the same company. Client information is also maintained by the company. Site information is separately maintained. Material information including name, amount available and reorder level is also to be maintained. Materials used for each construction are to be stored which include the name, model, type (like iron, steel, plastic etc.), quantity, company etc. Information of the retailer which provide these materials are also stored. Design a database system to enable the construction company to maintain all the data conveniently and efficiently.

Construction Table

Columns:

- Client-id
- Client-Name
- Client-Address
- Client-Contact
- House-id
- House-Address
- House-Type
- Site-id
- Site-Location
- Site-Area
- Material-id
- Material-Name
- Material-Model
- Material-Type
- Rec-lvl
- Qtyavail
- Company
- Retailer-id
- Retailer-Name
- Retailer-Address
- Retailer-Landline
- Retailer-Contact-no

Normalization Steps

Step 1: Identify Functional Dependencies

1. **Client Details:**
 - Client-id → Client-Name, Client-Address
 - Client-id → Contact-no
2. **House/Office:**
 - House-id → House-Address, House-Type, Client-id
3. **Site Details:**
 - Site-id → Site-Location, Site-Area, House-id
4. **Material Details:**
 - Material-id → Material-Name, Material-Model, Material-Type, Rec-lvl, Qtyavail, Company
5. **Retailer Details:**
 - Retailer-id → Retailer-Name, Retailer-Address, Landline-no
 - Retailer-id → Contact-no
6. **Relationships:**
 - Site-id ↔ Material-id (Uses table)
 - Material-id ↔ Retailer-id (Supplies table)

Step 2: Normalize to 1NF (First Normal Form)

1NF ensures:

- No repeating groups or multivalued attributes.
- Each cell contains atomic values.

In this table, multiple Contact-no values for Client-id or Retailer-id would violate 1NF. To resolve this:

1. Split Client-Contact into a separate table.
2. Split Retailer-Contact into a separate table.

Step 3: Normalize to 2NF (Second Normal Form)

2NF ensures:

- The table is in 1NF.
- No partial dependencies (i.e., non-key attributes depend on part of the primary key).

To resolve partial dependencies:

- **Client** details (Client-Name, Client-Address) depend on Client-id.
- **House/Office** details depend on House-id.
- **Site** details depend on Site-id.
- **Material** details depend on Material-id.
- **Retailer** details depend on Retailer-id.

Split these into separate tables:

- Client(Client-id, Name, Address)
- Client_Contact(Client-id, Contact-no)
- House/Office(House-id, Address, Type, Client-id)
- Site(Site-id, Location, Area, House-id)
- Material(Material-id, Name, Model, Type, Rec-lvl, Qtyavail, Company)
- Retailer(Retailer-id, Name, Address, Landline-no)
- Retailer_Contact(Retailer-id, Contact-no)

Step 4: Normalize to 3NF (Third Normal Form)

3NF ensures:

- The table is in 2NF.
- No transitive dependencies (i.e., non-key attributes depend on other non-key attributes).

At this point:

- All transitive dependencies have been removed.
- Each table contains attributes fully dependent on its primary key.

We also introduce tables for the relationships:

1. **Uses(Site-id, Material-id):** Relationship between Site and Material.
2. **Supplies(Material-id, Retailer-id):** Relationship between Material and Retailer.

Final Normalized Tables

Client (Client-id, Name, Address)

Client_Contact (Client-id, Contact-no)

House/Office (House-id, Address, Type, Client-id)

Site (Site-id, Location, Area, House-id)

Material (Material-id, Name, Model, Type, Rec-lvl, Qtyavail, Company)

Uses (Site-id, Material-id)

Retailer (Retailer-id, Name, Address, Landad-no)

Retailer_Contact (Retailer-id, Contact-no)

Supplies (Material-id, Retailer-id)